

AutoTheme: A Multi-Agent Framework for Inductive Thematic Analysis with LLMs

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Abstract

Thematic analysis is a qualitative research method used to identify and interpret patterns in textual data. However, it can be time-consuming and challenging to replicate. While recent advancements in large language models (LLMs) and generative AI have enhanced thematic analysis, existing methods often rely on prompt-based interactions and require significant human intervention. This paper introduces an agentic AI framework comprising autonomous, goal-directed agents powered by LLMs to perform inductive thematic analysis with minimal human input. We evaluate the framework using a dataset from a Cognitive Behavioral Therapy (CBT) mobile app and compare the results with those from Latent Dirichlet Allocation (LDA), demonstrating improved efficiency, adaptability, and thematic depth. Overall, the approach has shown efficacy, especially for short texts, such as app reviews and social media posts.

Keywords: AI agents, Inductive coding, CBT app review, Large language models.

1. Introduction

Thematic analysis is a qualitative research method that aims at identifying, analyzing, and reporting themes within textual data (Castleberry & Nolen, 2018; Skinner et al., 2022) to help understand people's experiences and perceptions about a phenomenon (Jowsey et al., 2021). Thematic analysis is widely used since it can help address a wide variety of research questions (Castleberry & Nolen, 2018). Furthermore, it is not tied to any theoretical framework and captures the essence of the data by systematically identifying the themes within data (Sandhiya & Bhuvaneshwari, 2025; Willig & Rogers, 2017). Thematic analysis is effective in identifying themes within textual data by allowing researchers to code and categorize data,

which in turn helps uncover relevant themes that describe the meanings and experiences of people (Braun & Clarke, 2021). This makes thematic analysis useful for exploring social phenomena, like people's attitudes, perceptions, beliefs, and behaviors, in different contexts and domains (Babar et al., 2018; Harper & Thompson, 2011; Kiger & and Varpio, 2020; Oyeboode et al., 2021).

Inductive thematic analysis is used to identify, analyze, and interpret patterns within textual data. It involves coding and developing themes based on the content of the data, without any predefined categories, which allows themes to emerge from the users' perspectives and experiences (De Paoli, 2024).

Despite the popularity of thematic analysis, it suffers from several inherent challenges and limitations. Precisely defining themes using thematic analysis is sometimes challenging, making it difficult for others to replicate findings (King et al., 2018). In addition, thematic analysis may produce overly simplistic themes by overlooking nuanced meanings embedded within data (Vaismoradi et al., 2013). Finally, thematic analysis is time-consuming in terms of coding, categorizing, and refining themes, especially with large datasets (Maguire & Delahunt, 2017).

Large Language Models (LLMs) have advanced the field of natural language processing (NLP) and have demonstrated promising capabilities in various tasks. LLMs can understand text and perform tasks related to sentiment analysis, named entity recognition, and text summarization (W. Khan et al., 2023; Min et al., 2023). LLMs are also capable of generating precise answers to different questions by understanding the context and extracting relevant information from large datasets (Jiang et al., 2021; Singhal et al., 2023). In addition, LLMs can generate human-like text (Orrù et al., 2023), coherent narrative and dialogue (Wang, Fu, et al., 2025), personalize

content (Haag et al., 2025), and generate tailored responses (Yalamanchili et al., 2024).

The integration of LLMs into thematic analysis can enhance efficiency, accuracy, and depth of analysis. LLMs can comprehend human-generated content (Nasution & Onan, 2024), making them valuable tools across various domains, including qualitative research (Hamilton et al., 2023; Joel-Edgar & Pan, 2024). LLMs can enhance the efficiency of thematic analysis by automating the process, which requires substantial time and effort when done manually (Carvalho et al., 2024; Zhang et al., 2024). Furthermore, LLMs can help improve the thematic analysis process by uncovering nuanced themes that human analysts might overlook (Zhang et al., 2024; Zhao et al., 2024).

AI agents are autonomous systems designed to perform tasks with minimal human intervention, leveraging advanced computational models to make decisions, learn from environments, and adapt to new situations. Agentic AI refers to systems designed to pursue complex goals autonomously with minimal human intervention (Acharya et al., 2025). Agents are guided by LLMs and have revolutionized the way tasks are completed in different domains (Deng et al., 2025). AI agents can be used to analyze qualitative data through pattern identification and detection (Acharya et al., 2025) by leveraging LLMs' advanced capabilities in reasoning and performing several tasks more efficiently and effectively (Deng et al., 2025).

In this paper, we aim to address some of the limitations associated with thematic analysis when used with large datasets. In addition, we aim to use agents in thematic analysis without human intervention and eliminate the need for prompt-based interactions to identify themes from the data. Finally, we investigate the potential of using agentic AI with short texts with limited context windows.

The remainder of this paper is organized as follows: Section 2 provides a comprehensive review of the relevant literature on the use of LLMs, GenAI, and AI agents in thematic analysis. Section three provides an overview of the methodology used to conduct thematic analysis using AI agents. Section four presents empirical findings from applying our multi-agent workflow to a dataset of a Cognitive Behavioral Therapy (CBT) mobile app. Section five discusses these results in relation to a topic analysis-based approach using Latent Dirichlet Allocation (LDA). Finally, section six concludes the paper by summarizing our key contributions and acknowledging any limitations.

2. Literature Review

According to the literature, several studies have demonstrated how AI, GenAI, LLMs, and AI agents can be used to complement traditional thematic analysis approaches. For example, Zhang et al., (2025) explored the potential of ChatGPT in thematic analysis as well as the associated concerns and challenges related to integrating ChatGPT into qualitative analysis. The authors proposed a six-phase framework for designing prompts through collaboration with qualitative researchers. The authors used a transcript of focus groups to be analyzed using ChatGPT as well as expert qualitative researchers. Results showed that ChatGPT has the potential to enhance thematic analysis and reduce the time and effort required for this process. Furthermore, the participants stated that improved transparency and guidance on prompts can enhance collaboration with ChatGPT.

Khan et al., (2024) explored the use of LLMs in supporting thematic analysis of controversial topics. The authors used a combination of human coders and LLMs (GPT-4 and Llama 2) to analyze social media posts and government-related reports. Human coders generated 11 themes that were used to compare the initial analysis by the LLMs. Based on the initial comparison, the authors revised the prompts, and the analysis was repeated using updated models (GPT-4, Claude 3 Opus, and Llama 3). The process involved iterative refinement of prompts and thematic coding, resulting in a final set of themes derived from both the LLMs and human coders.

Perkins & Roe, (2024) used GenAI as a tool to assist researchers in thematic analysis. Using a semantic-level thematic analysis framework and realist epistemology, two researchers analyzed academic policies on GenAI, one manually and the other using ChatGPT. An iterative prompt technique was employed to address challenges in obtaining the desired outcomes. Results showed that the ChatGPT-based approach enables a richer and more nuanced understanding of the themes present in the data. The ChatGPT-based approach offered speed, granularity, and efficiency. On the other hand, the manual analysis approach provided greater contextual depth and nuance. Katz et al., (2024) explored the role of open-source generative text models in thematic analysis and proposed a workflow that is based on machine learning, natural language processing tools, and generative text models to create codebooks inductively. The authors evaluated the proposed workflow using three synthetic datasets. Results showed that the proposed workflow could generate themes closely matching original sub-themes in synthetic datasets.

Turobov et al., (2024) explored the use of ChatGPT in the thematic analysis of UN policy documents. The authors utilized ChatGPT to generate themes from the data through prompt engineering and verification processes and then validated the outcomes using LDA-based topic modeling. Results showed that ChatGPT effectively generated descriptive and thematically relevant codes. Furthermore, it increases productivity and enhances quality through comprehensive analysis of complex datasets. Deiner et al., (2024) assessed the utility of LLMs in inductive thematic analysis using social media data. Compared to previously published findings, results showed that LLMs can generate the top themes identified manually by human experts. Prescott et al., (2024) compared the performance of ChatGPT and Bart in conducting thematic analysis with human experts. The authors used a set of text messages to promote patients' adherence to their antiretroviral medication. Results showed that both LLMs consistently generated about 71% of the themes identified by human experts using inductive analysis. On the other hand, the deductive analysis yielded 50% agreement between ChatGPT and human experts, and 58% agreement between Bard and human subjects.

Wang et al., (2025) evaluated LLM-based thematic analysis using GPT-4 and Gemini. Using semi-structured interviews with "researchers to gather insights on their experiences generating and analyzing Online Social Network (OSN) communications datasets". The authors initially analyzed the dataset using Braun and Clarke's six-phase thematic analysis framework. Prompt engineering was used to design prompts to guide thematic analysis using LLMs, and the findings were compared with results from manual analysis. Results showed that LLMs achieved a cosine similarity of 0.76, indicating good performance of LLMs in conducting thematic analysis. Yang et al., (2024) evaluated the use of the GPT-4 model in the thematic analysis of interview transcripts that were tokenized with a 100-token overlap. The authors designed the prompt with the needed context and desired output. Results showed that GPT-4 was able to identify at least 50% of the themes generated by human experts.

Amani et al., (2025) evaluated the reliability of ChatGPT inductive and deductive thematic coding compared to traditional manual methods. The authors used two datasets: one for inductive coding to generate themes, and the other for deductive analysis using a predefined codebook. Results showed that ChatGPT enhanced the efficiency of thematic analysis but still requires human oversight for accuracy and ethical integrity. Gamielien et al., (2023) compared traditional thematic analysis techniques with LLMs,

more specifically, NLP Cluster Assisted (NLPCA) and NLP with GPT-3.5 (NLPGPT) for analyzing qualitative data that consists of 3,800 structured reflections prompting students to self-reflect after receiving graded exams. Results showed that NLPCA and NLPGPT methods identified similar themes, indicating the capabilities of LLMs to achieve similar performance of manual thematic analysis.

S. Qiao et al., (2025) explored the use of GenAI to enhance the process of thematic analysis. Using transcripts from semi-structured interviews with maternity care providers conducted during the COVID-19 pandemic, the authors used ChatGPT to perform thematic analysis using a well-defined pipeline and compared the results against human-generated themes. Results showed that ChatGPT achieved 80% accuracy in generating codes from interview transcripts while enhancing the efficiency in thematic analysis. Breazu et al., (2024) explored the use of LLMs in inductive and deductive analysis. The authors used a YouTube comments dataset that was analyzed by a human expert using NVivo. Results from the LLMs were compared with those from NVivo, and the findings showed that the LLM results resemble those of a human expert using NVivo.

Flanders et al., (2025) explored the use of ChatGPT in the thematic analysis of Malaysian news articles on refugees. The authors used combinations of few-shot prompting, clustering, and a multi-step Socratic prompting approach. Results showed that ChatGPT achieved an F1 score of 82%, with high recall and precision in identifying irrelevant text. Sabbaghan, (2024) explored the use of TRACER, a GenAI-driven tool, in thematic analysis. The author employed a mixed-methods comparative data analysis to evaluate human and TRACER performance, using interview transcripts from educators implementing a new mathematics instructional model. Results showed that TRACER's analysis revealed valuable insights but was not fully aligned with the themes identified by human experts.

Xu et al., (2025) explored the use of a human-AI collaborative thematic analysis framework for clinical interviews. The framework consists of a multi-agent system that is based on structured conversations between agents and the expertise of cardiac experts in thematic analysis. The authors evaluated the proposed framework using interview transcripts from parents of children with rare congenital heart disease. Results showed that the proposed framework outperformed existing LLM-assisted thematic approaches, achieving higher thematic hit rates and coverage.

Sankaranarayanan et al., (2025) proposed a thematic analysis approach based on an orchestrator LLM agent and a set of multiple LLM sub-agents

reflecting all steps in traditional manual thematic analysis. The authors evaluated the proposed approach using expert statements about computer-supported collaborative learning. Results showed that the proposed approach had strong robustness, transparency, and validity, producing consistent results aligned with manual themes. Furthermore, it reduced time and effort while maintaining clarity.

T. Qiao et al., (2025) proposed a multi-agent system for thematic analysis that consists of several agents for coding, code aggregation, and codebook maintenance and update. The authors tested the system using Dreddit dataset and the Reddit climate change dataset. Results showed that the system was able to find eight themes and was able to address limitations associated with traditional thematic analysis approaches.

While several studies demonstrated LLMs, like ChatGPT, Llama, and Gemini, could enhance the efficiency and effectiveness of thematic analysis (Amani et al., 2025; Sabbaghan, 2024; Turobov et al., 2024), these approaches focus on prompt-based interactions. Such approaches generally require ongoing human intervention to guide, refine, and validate the analysis process (Amani et al., 2025; Prescott et al., 2024; Xu et al., 2025; Yan et al., 2024; Zhang et al., 2025).

In addition, current works focus on relatively long text, taking advantage of the broader context window supported by LLMs models. This concentration on long-form text overlooked the challenges associated with a short text, such as users' reviews or social media posts, which suffer from limited contextual information and, therefore, could be challenging in tasks like the inductive thematic analysis (Flanders et al., 2025; Katz et al., 2024; Nguyen-Trung, 2025; Perkins & Roe, 2024; Turobov et al., 2024; Wang, Fu, et al., 2025).

Furthermore, current studies have not explored the potential of autonomous, goal-directed agents, known as agentic AI systems, that perform tasks with minimal human intervention to facilitate inductive thematic analysis. Using an agentic AI system to perform inductive thematic analysis could help dynamically manage the entire process through iterative refinement and adaptation to varying text lengths, including short texts. Such agentic AI systems could have the ability to overcome limitations inherent in static, prompt-driven approaches, especially when it comes to short texts with limited context.

3. Methodology

The methodology for conducting a thematic analysis of user reviews is illustrated in Figure 1. The

methodology begins with a set of user reviews collected from the Play Store for one of the popular CBT mobile apps.

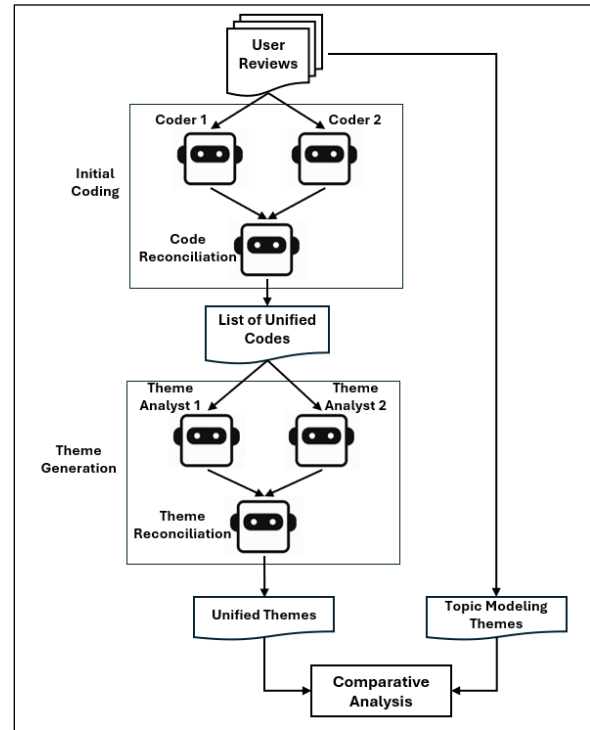


Figure 1. Multi-Agent AI thematic analysis methodology.

For the subsequent steps, namely initial coding and theme generation, we utilized the OpenAI agents SDK to develop an multi-agent application that performs the basic steps of thematic analysis (*OpenAI Agents SDK*, 2025) using GPT-4.1 as the model. Tables 1, 2, 3, and 4 show the agents' prompts used in the agentic application for automatic thematic analysis.

Table 1. Coder Agents' prompts.

Coder1_Agent_Prompt = ("You are a qualitative research assistant performing initial coding on user reviews for a CBT mobile application. For each review, extract only the features, qualities, or specific aspects of the app that the reviewer explicitly mentions as making them satisfied with the app. Only focus on functional aspects of the app. Ignore anything non-technical like subscription, security, etc. Do NOT infer or assume satisfaction where it is not stated. Return a comma-separated list of such codes if more than one is available. Example: For 'I love how easy it is to track my moods', the code would be 'mood tracking, easy to use'.")

Coder2_Agent_Prompt = Coder1_Prompt

Table 2. Code Reconciler Agent' prompts.

Reconcile_Agent_Prompt = (
"You are supervising qualitative initial coding for CBT app user reviews. You receive two lists of initial codes from two coders. Merge them, remove duplicates, and ensure each code reflects only what is directly mentioned in the review. Do not generate any codes other than the one entered by the two coders. If both initial coders assigned NA or did not state any codes, then simply list the final codes as NA. Return a comma-separated list of codes.")

Table 3. Theme Generator Agents' prompts.

Theme_Generator1_Agent_Prompt = "You are an expert in qualitative research performing thematic analysis on a collection of initial codes. Your task is to group related codes into broader themes reflecting design features. While grouping, ignore any codes that state 'No codes available', 'No explicit functional aspects mentioned', 'No explicit aspects mentioned', etc... Ignore non-technical/non-functional design features like simple interface, accessibility, free cost, etc... Focus the themes on what the app actually offers to support therapeutic outcomes. Output as: Theme: Label \n- code1 \n- code2 \nOnly group codes based on explicit similarity."

Theme_Generator2_Agent_Prompt =
Theme_Generator1_Prompt

Table 4. Theme Reconciler Agent' prompts.

Theme_Reconcile_Agent_Prompt = ("You are supervising thematic analysis. You have two separate set of themes with a set of codes under each theme. Reconcile the themes from both sets into a unified set and remove duplicates. If needed, merge related themes into a unified theme reflecting the merged themes. Output as: Theme: Label \n- code1 \n- code2.")

A Streamlit (Khorasani et al., 2022) Web application was developed using Python. This Streamlit application is designed to perform initial coding and thematic analysis on user data. The application is a pipeline that leverages checkpoints during the processing of user data. It ensures that progress is saved during long or interrupted coding sessions, minimizing redundant work. In addition, the Streamlit app supports monitoring, validation, and reuse of the autonomous multi-agent workflow across different datasets. The application employs multi-agent logic, where multiple AI agents simulate a

collaborative coding process. Two agents independently code each review, and a third agent reconciles their outputs, emulating the practice of inter-rater reliability in qualitative analysis. Similar logic is applied for thematics analysis.

The initial coding process consists of three agents. Two independent AI agents, each systematically extracting explicit design features that contribute to users' satisfaction with the CBT app. Next, an AI agent reconciled discrepancies between the two independently generated coding outputs, ensuring duplicates were removed and all codes explicitly reflected the original reviews.

The theme generation step starts with a consolidated set of unique initial codes and three agents. The first two agents performed separate thematic analyses and independently grouped the initial codes into broader, meaningful themes based solely on explicit similarity. The third agent, namely the theme reconciler, reconciled the thematic groupings, resolving differences and merging them into a unified thematic structure.

The final set of themes represents the design features of CBT applications that contribute to user satisfaction with these applications. The final set of themes generated from the agents was compared to a set of themes generated using Latent Dirichlet Allocation (LDA) and PyLDAvis visualization techniques on the same data set(El-Gayar et al., 2024). The LDA results were labeled by two independent researchers, with a Cohen's Kappa statistics of 0.86, which reflects substantial agreement among the raters (Landis & Koch, 1977).

4. Results

We collected a total of 24,902 4 and 5-star rating reviews between April 2020 and April 2023. The 4- and 5-star rating reviews reflect a high level of satisfaction with the CBT mobile app and help extract themes that highlight functional aspects and therapeutic features associated with user satisfaction.

Initial coding resulted in a total of 20,062 codes based on the reconciliation agent. Appendix A shows sample coded reviews from the initial coding step using the three agents. Coder 1 was able to analyze a total of 4,847 reviews, Coder 2 was able to analyze a total of 4,781 reviews, the Code Reconciliation was able to unify a total of 5,675 reviews. The difference between Coder 1 and Coder 2 analysis is attributed to the fact that for some reviews, Coder 1 could identify initial codes while Coder 2 did not.

Table 5 shows the results from the theme generation step. Theme Analyst 1 and Theme Analyst 2 were able to identify 14 and 13 design feature

themes, respectively. The Theme Reconciliation agent was able to identify a total of 14 design feature themes.

Table 5. Themes identified by the three agents.

Theme Analyst 1	Theme Analyst 2	Theme Reconciliation
Personality Assessment & Insights	Personality Assessment & Insights	CBT, DBT & Psychological Tools
Self-Reflection, Journaling & Tracking	Customization & Personalization	Emotional Awareness, Processing & Regulation
CBT, Therapeutic Tools & Psychological Techniques	Journaling & Self-Reflection Tools	Personality Assessment & Insights
Guided Journeys, Learning Paths, and Structured Programs	Rescue/ Emergency Sessions	Progress Monitoring, Feedback & Tracking
Rescue & Emergency Support Features	Question-Based Guidance & Assessment	Question-Based & Interactive Guidance
Customization, Personalization, and Adaptive Content	CBT, DBT & Therapeutic Approaches	Actionable Steps, Practical Tools & Goal Setting
Actionable Steps, Practical Tools & Goal Setting	Behavioral & Habits Change Support	Rescue & Emergency Support Features
Reflection, Insight & Awareness	Emotional Awareness, Processing & Regulation	Short, Manageable & Bite-Sized Sessions
Habit & Routine Support	Learning Paths, Courses & Content Structure	Learning Paths, Courses & Structured Programs
Problem Identification & Issue Management	Progress, Goal setting & Tracking	Behavioral Change, Habits & Routine Support
Short, Bite-Sized Sessions & Digestibility	Guidance, Prompts & Support	Journaling & Self-Reflection Tools
Emotional Support & Regulation	Short, Manageable, & Bite-Sized Sessions	Insight, Self-Awareness & Reflection
Progress Monitoring & Feedback	Tracking and Monitoring Tools	Customization & Personalization
AI, Automation & Digital Support		AI, Automation & Digital Support

Table 6 shows the results from the theme generation step and those obtained using LDA topic mining. The Theme Reconciliation agent was able to identify a total of 10 design feature themes, while the LDA-based approach was able to identify a total of 6 design feature themes.

Table 6. Final design feature themes identified using the agentic-based approach and LDA-based approach.

Agentic Based Design Features	LDA Based Design Features
CBT, DBT & Psychological Tools	Mental Health Management and Support
Emotional Awareness, Processing & Regulation	
Personality Assessment & Insights	Credibility Support (boost trust, expertise, and reliability (e.g., assessments, accuracy))
Progress Monitoring, Feedback & Tracking	
Question-Based & Interactive Guidance	Therapeutic Approaches and Tools
Actionable Steps, Practical Tools & Goal Setting	
Rescue & Emergency Support Features	Beneficial Rescue Sessions
Short, Manageable & Bite-Sized Sessions	
Learning Paths, Courses & Structured Programs	Personal Growth and Development
Behavioral Change, Habits & Routine Support	
Journaling & Self-Reflection Tools	Self-Understanding and Personality Insights
Insight, Self-Awareness & Reflection	
Customization & Personalization	
AI, Automation & Digital Support	

5. Discussion

The reconciled themes emphasized personalized user experiences, accurate assessments, mindfulness practices, functionality, educational content, specific CBT tools, journaling, and self-reflection. This demonstrates the effectiveness of Agentic AI approaches in extracting nuanced insights from datasets comprising short user reviews.

Comparing agentic-based design feature themes with those derived from LDA-based topic modeling, we can notice that the Agentic AI approach generated more detailed and comprehensive design feature themes, consisting of 14 distinct themes, such as customization, emotional regulation tools, personality insights, actionable guidance, rescue features,

structured learning paths, and behavioral change support. On the other hand, the LDA-based approach yielded six broader, aggregated themes: mental health management, credibility support, therapeutic approaches, beneficial rescue sessions, personal growth, and self-understanding.

Comparing the generated design feature themes from both approaches as well as the mapping shown in table 3, we can notice how the agentic AI approach achieves higher specificity and nuance, potentially providing more precise and detailed features that can enhance the mobile app significantly. In contrast, LDA provided broader, overarching thematic categories more suited to high-level design features of CBT apps.

The difference in terms of finding using the two approaches highlights their complementary nature. The agentic AI method provides detailed, actionable insights that are more geared towards guiding precise app development and feature refinement, emphasizing user-centered enhancements like personalized interactions, emotional and psychological support mechanisms, and structured, goal-oriented content. On the other hand, the LDA-based approach is better suited to provide broad thematic trends, which in turn capture the overall user experience with CBT apps. Accordingly, using both approaches could significantly enhance the depth and breadth of insights from user-generated reviews, which in turn provide high-level as well as low-level design feature themes that could ultimately enhance the design and development of CBT apps.

Finally, results demonstrate the effectiveness of the agentic AI approach in thematic analysis for short text, given the detailed design features identified by the agents and how they compare to the design features from traditional approaches like LDA that require human intervention for labeling topics.

6. Conclusion

In this paper, we presented an approach for leveraging AI agents for inductive thematic analysis. We demonstrated the proposed approach by analyzing online-generated user reviews of a popular CBT mobile application to identify design features that contribute to a positive user experience with the app. We compared the findings with design features generated using the well-known LDA topic modeling technique.

Overall, the findings underscore the methodological strengths of employing Agentic AI for nuanced thematic analysis of user-generated content, demonstrating the method's capability to uncover detailed insights essential to improving CBT apps. On the other hand, the LDA-based approach effectively

provides high-level themes, highlighting the complementary roles the two approaches can play in thematic analysis in user experience research.

This work is not without limitations. First, the approach depends on a single LLM (ChatGPT), which could impact the generalizability of the findings. Second, using multiple agents per review and the reconciliation agent could increase computational resources and time, potentially limiting scalability to larger datasets.

Future work could address these limitations in several ways. First, comparing the framework against additional state-of-the-art LLMs (e.g., Claude, Gemini, or Llama) could provide a stronger benchmark of performance. Second, analyzing the computational costs of the multi-agent workflow and examining how errors may propagate across agents are important directions for enhancing scalability, robustness, and reliability in large-scale applications. Third, exploring the impact of alternative prompt formulations on the number and nature of codes and themes extracted could provide further insights into how prompt design influences the outcomes of thematic analysis. Fourth, expanding the analysis to include lower-rated reviews could yield complementary insights by highlighting usability challenges and areas where CBT apps may fall short in meeting user expectations. Finally, quantifying the distribution of generated themes and analyzing their frequency would help highlight the most prominent design features and improve interpretability.

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Appendix A: Sample Coded Reviews Using Three Agents

Review	Coder 1	Coder 2	Code Reconciliation
Just started this app and I'm amazed at how accurately this describes me and how insightful it has already been. I'm looking forward to diving in to this!	accurate descriptions, insightful feedback	accurate insights	accurate descriptions, insightful feedback, accurate insights
Very good analysis reports. So far the app has really impressed me.	analysis reports	analysis reports	analysis reports
The choices are diverse ,which allows me not to stress my self and somehow helpful.	diverse choices	diverse choices	diverse choices
It's a great app to help us in emocional intelligence.	emotional intelligence	emotional intelligence	emotional intelligence
Amazing app so far, a real eye opener. It allows you to reflect on your own emotional state, it makes it easy to do so.	emotional reflection, easy to use	emotional reflection, easy to use	emotional reflection, easy to use
Best self help aap Best explanation for your ans as well tell you everything about you	explanation, personalized insights	explanations, personalized insights	explanation, personalized insights
Simply brilliant! I end up figuring out what my problems are even before I finish writing my journal(s). The ability to access your answers later is also a wonderful touch. If I ever get the chance to, I'd definitely recommend this app 10/10.	journaling feature, access to past answers	journal feature, access to saved answers	journaling feature, access to past answers
I am appreciative of the journalling aspect of this app. Many themes which lead the user through step by step. The meditation are pleasant too.	journalling, step-by-step themes, pleasant meditation	journaling, guided themes, meditation	journalling, step-by-step themes, pleasant meditation, guided themes
Helps in being mindful and gives you understanding of the why of mental conditions or states and helps you toward better habits.	mindfulness, understanding mental conditions, habit formation	mindfulness support, understanding mental conditions, habit formation	mindfulness, understanding mental conditions, habit formation
Exceptional! The app really reflects how well it understands human behaviour.	understands human behavior	none	understands human behavior